Nathan Nicholas D'Souza

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SKILLS & INTERESTS

- **AI Engineering & Machine Learning:** LangGraph, Hybrid RAG (BM25 + Dense RRF), Semantic Caching, LLM API Orchestration (Claude API), Agentic Workflows (MCPs, Connectors), Guardrails & Eval Harnesses, Quantized ONNX (FastEmbed), ONNX Runtime / DirectML, PyTorch, TensorFlow, Keras, OpenCV, Pytesseract (OCR), Computer Vision.
- **Languages & Desktop/Systems:** Python (FastAPI, Django, Flask, PySide6), C# / .NET 10, WPF (MVVM), Win32 Interop (P/Invoke), WebView2, WebGL2 / GLSL, Audio Signal Processing (FFT), TypeScript & JavaScript (Next.js, React, React Native).
- **MLOps, Data & Infrastructure:** Docker, CI/CD (GitHub Actions), AWS (Lambda, API Gateway, DynamoDB), MSIX & Microsoft Store Packaging, PostgreSQL (pgvector), Redis, SQL, ETL Pipelines, PyTest, xUnit, Vitest, Playwright, Power BI, Pandas, NumPy, Agile/Scrum (Jira).

EXPERIENCE

Assistant, HR Consulting Services | *Fraser Health Authority - Surrey, Canada* *Sept 2025 – Present*

- **Architected full-stack SDLC solutions** using React Native/Django/SQLite to implement digital phase-tracking and automated paper trails for 100% of agency contracts.
- **Developed ETL-driven dashboards** with Django and Chart.js to visualize 8-week staffing trends, eliminating high-latency manual data retrieval from Outlook.
- **Optimized legacy automation suites** in PySide6, updating script logic to ensure 100% reliability for critical departmental workflows and real-time data accessibility.

Workforce QI and Optimization Co-op | *Fraser Health Authority - Surrey, Canada* *Aug 2024 – Aug 2025*

- **Developed ETL pipelines** via a PySide6/pywin32 RFP Utility to automate invoice extraction and organization, reducing weekly manual labor from **30+ hours to under 5 minutes.**
- **Accelerated UKG WFM migration by 60%** through Python-based audit automation and custom PDF-to-Excel statement reporting for complex vendor payment tracking.
- **Engineered Power BI analytics** for workforce scheduling and payroll, performing regression-style checks to ensure data integrity and equitable employee shift distribution.

PROJECTS

Wavelit | *Live Wallpaper App for Windows (C#, .NET 10, WPF, WebGL2, Win32)* | *Link* *2026 - Present*

- **Designed and built a Windows desktop app for the Microsoft Store** (~23,000 lines of C#, JS, XAML) with **240+ automated unit tests** and a GitHub Actions pipeline that builds, tests, and packages the MSIX per push.
- **Built a wallpaper engine using Win32 programming** that draws behind desktop icons on Windows 10 and 11 (incl. 24H2), across multiple monitors at different scales, with a shared-canvas "extend" mode.
- **Built power-aware playback** that suspends rendering when covered (GPU near zero) and an adaptive quality guard, plus a WebGL2 shader runtime with 19 original GLSL shaders and WASAPI/FFT music visualizers.
- **Built an AI shader Studio** on the Claude API (structured output, prompt caching, fallback) where every AI shader passes a static check, compile, and timed GPU benchmark before use, plus on-device SDXL-Lightning image generation (ONNX Runtime/DirectML) with a safety classifier.

DualSight (formerly Aegis AI Race) | *Engineered AI vs. Naive Chatbot* | *Link* *Sept 2026 - Present*

- **Engineered an agentic Hybrid RAG pipeline** (FastAPI, LangGraph, PostgreSQL pgvector + BM25 via RRF, Redis semantic cache) scoring **9/10 vs 3/10** correct on a live-model audit trap, with 7.3 ms cache hits and 28 ms retrieval.
- **Validated with ~350 automated tests** and evaluation harnesses (retrieval ablation, cache-threshold sweep, live accuracy) in a GitHub Actions CI pipeline; deployed on Vercel and Render (Docker).

Armrest | *Phone Voice & Touch Remote for a Windows PC* | *Link* *2026 - Present*

- **Built an AWS serverless relay** (Lambda, API Gateway, DynamoDB), Python Windows agent, and React phone app.

EDUCATION

Simon Fraser University | *BSc in Computing Science, Conc. Artificial Intelligence* *Sept 2020 – Dec 2025*

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Database Systems, Computer Vision, Data Structures & Algorithms, Software Engineering.

AWARDS & ACHIEVEMENTS

Mountain Madness 2025: Placed **1st** as "The Classic" Hackathon Winners, earning CAD **200** *March 2025*

Opportunity Fest 2023: Awarded CAD **250** for **2nd** place in **Social Innovation** as HelpingHands *November 2023*